EMPLOYEE ATTRITION AND DEPARTMENT PERFORMANCE ANALYSIS

1. PROJECT OVERVIEW:

- This project focuses on analysing Human Resources data related to employee diversity, performance, salary, attrition and understand employee retention patterns and evaluate department-wise performance
- The project "Employee Attrition and Department Performance Analysis" contains detailed information about key HR attributes such as employee age, gender, department, job level, salary, performance rating, promotion status, and attrition information.
- The data is structured to provide insights into employee retention patterns and to evaluate performance variations across different departments and job roles.
- The main objective of this project is to demonstrate employee retention, and department-wise performance for better HR decision-making.

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2. DATA SOURCES:

Source description and Timeline: The dataset was obtained from Google Dataset Search (HR Analytics / Diversity Dataset) and 11-04-2025

Domain: Human Resources

Source link:

https://figshare.com/articles/dataset/Human_Resources_csv/28780886?file=53616341

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3. PROBLEM STATEMENT:

- To analyse employee attrition and retention patterns.
- To compare department-wise salary and performance.
- To evaluate workforce diversity across gender and job levels.

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4. ATTRIBUTE (COLUMN/FEATURES) DETAILS:

Attribute Name	Data Type	Description
Employee-ID	Numeric	Unique identifier for each employee
Gender	Text	Employee gender
Age	Number	Age of the employee
Department	Text	Department of employee
Birth_date	Date	Employee date of birth
Hire_date	Date	Employee joining date
Race	Text	Ethnic classification of employee
Age_group	Categorical	Categorized age group
Experience_years	Date	Years of experience calculated from hire date
Experience_Group	Numeric	Categorized experience level
Attrition	Numeric	Indicates whether employee left(1) or stay(0)

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5 .TOOLS AND TECHNOLOGIES:

- **Excel:** Data cleaning, transformation, and pivot tables.
- **Power BI:** Data modelling, DAX calculations, visualization, and interactive dashboard creation.

6.DATA PRE-PROCESSING:

Tasks Performed:

- **Data Cleaning:**
  Removed duplicate records from the dataset. Checked all columns for blank and missing values. Standardized date format for Birth_Date and Hire_Date columns. Converted the

dataset into an Excel Table for structured analysis. Ensured consistency in categorical columns such as Department, Gender, and Race.

- **Transformation:**

  1.Termination date had blank values but blank termination date was treated as “Employee Active”. No major missing values were found in important columns like Department and Hire_Date.
  Formula: =IF(ISBLANK(TERMINATION_DATE),”ACTIVE”,”TERMINATED”)

  2.The new columns were created for analysis (i.e) Experience_group, Age_group and Attrition.

  Formula:

  =IF(employee_status=”TERMINATED”,1,0)

  =IF(age<=25,”18-25”,IF(age<=35,”26-35”,IF(age<=45,”36-45”,”46+”)))

  =IF(experience<=5,”0-5”,IF(experience<=15,”6-15”,IF(experience<=26,”16-26”)))

- **Pivot Tables:**

  The following Pivot Tables were created to analyse the data are Total Employees by Department, Attrition Count by Department, Attrition by Age Group, Attrition by Experience Group, Gender-wise Attrition. These Pivot Tables helped in performing descriptive analysis and identifying patterns in employee attrition.

- **Converted the data into Fact and Dimensional table.**

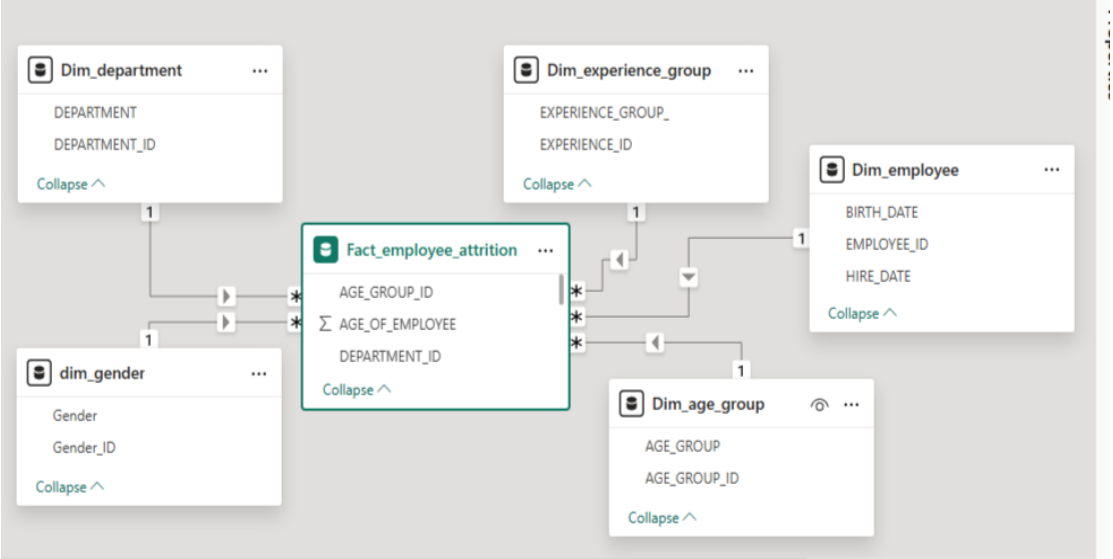
  1. Fact_Employee_attrition: Employee_ID, Department_ID, Gender_ID, Age_group_ID, Experience_group_ID, Age, Experience, Attrition.

  2. Dimenional_table: Dim_Department, Dim_Gender, Dim_Age_group, Dim_Experience_group, Dim_employee

  3. XLOOKUP function was used to fetch surrogate keys from dimensional tables into fact tablesamd ensure accurate mapping between descriptive attributes and surrogate IDs in the fact table.
  Formula:
  =XLOOKUP(Gender,Dim_gender!A:A,Dim_Gender!B:B)

7.DATA MODELLING AND DAX (POWER BI):

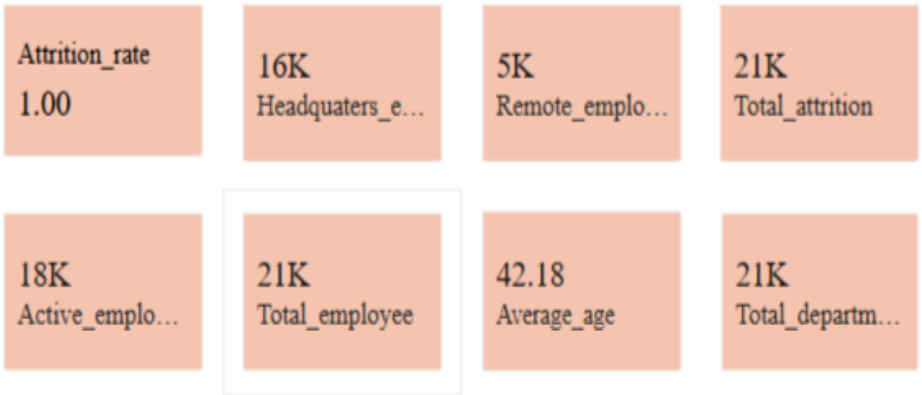
Data Model: Established relationships between tables, defined cardinality, and created lookup tables where necessary.



- A **Star Schema** data model was implemented with a central fact table (Fact_employee_attrition) connected to multiple dimension tables such as Department, Employee, Age Group, Experience Group, and Gender using one-to-many relationships.
- This modelling approach improves performance, ensures data integrity, and supports efficient analytical reporting in Power BI.
- **Fact_employee_attrition** contains transactional employee-level data. It stores measurable and analytical fields that includes foreign keys connecting to dimension tables. This table acts as the central table for analysis.
- **Dim_department** contains department details. DEPARTMENT_ID is the primary key that used for department-level analysis.
- **Dim_employee** contains employee master information. EMPLOYEE_ID is the primary key that includes Birth Date and Hire Date.
- **Dim_age_group** categorizes employees into age ranges. AGE_GROUP_ID is the primary key that used for demographic analysis.
- **Dim_experience_group** categorizes employees based on years of experience. EXPERIENCE_ID is the primary key that used for tenure-based attrition analysis.
- **Dim_gender** contains gender classification. Gender_ID is the primary key that used for gender-based workforce analysis.
- **All relationships are One-to-Many** (1:*) i.e., Dimension table (1 side) and Fact table (* side). Relationship Direction is Single direction filtering (Dimension → Fact) and all relationships are marked as active
- In cardinality, dimension tables contain unique values (Primary Keys). Fact table contains repeated foreign key values. This ensures proper data aggregation and avoids duplication.

DAX Measures: Implemented DAX formulas for key metrics

- Active_employees=CALCULATE(COUNTROWS(Fact_employee_attrition),Fact_employee_attrition[EMPLOYEE_STATUS]="ACTIVE")
- Attrition_rate = DIVIDE('Table'[Total_attrition],'Table'[Total_employee],0)
- Average_age = AVERAGE(Fact_employee_attrition[AGE_OF_EMPLOYEE])
- Headquaters_employee = CALCULATE(count('Human Resources'[WORK_LOCATION]),'Human Resources'[WORK_LOCATION]="Headquarters")
- Remote_employee = CALCULATE(COUNT('Human Resources'[WORK_LOCATION]),'Human Resources'[WORK_LOCATION]="Remote")
- Total_attrition = COUNT(Fact_employee_attrition[EMPLOYEE_ATTRITION])
- Total_department = COUNT(Fact_employee_attrition[DEPARTMENT_ID])
- Total_employee = COUNT(Fact_employee_attrition[EMPLOYEE_ID])
- Age_group = IF('Human Resources'[AGE_OF_EMPLOYEE]<30,"Young",IF('Human Resources'[AGE_OF_EMPLOYEE]<=40,"Mid Age","Senior"))



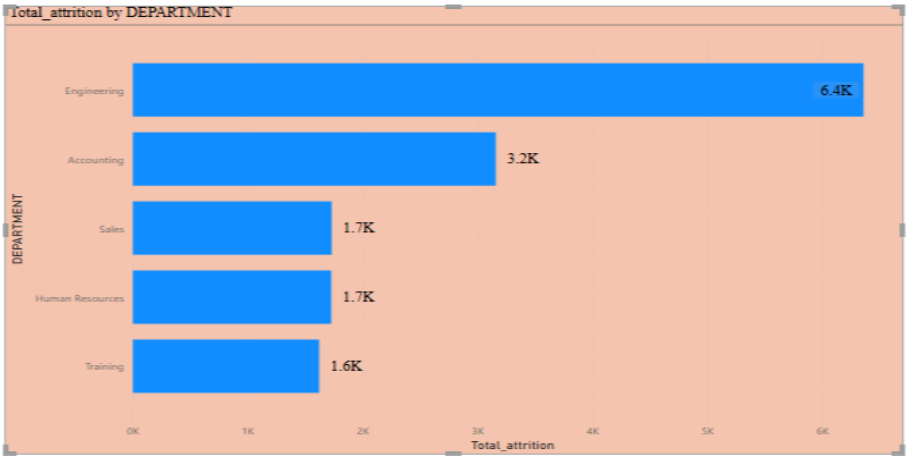
8. ANALYSIS AND VISUALIZATIONS (POWER BI):

Visualizations based on problem statement:

1.Attrition by Department

Insights:

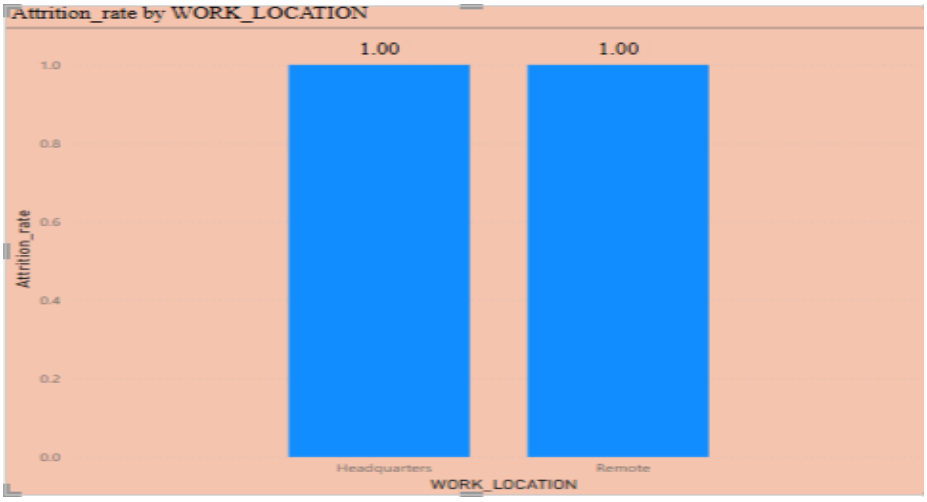
Identifies departments with highest employee turnover. Highlights departments that need retention strategies. Helps management focus on high-risk areas.



2. Attrition by Work Location

Insights:

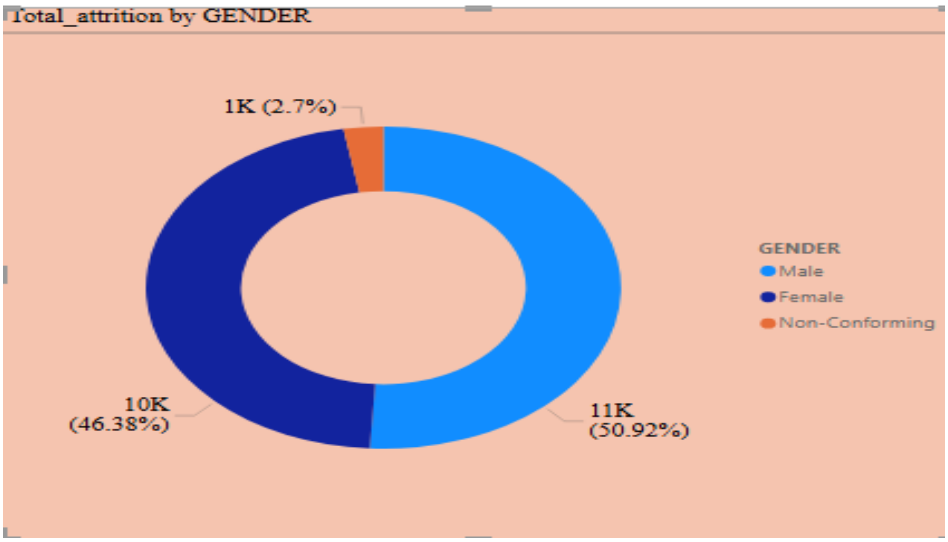
Compares attrition between Remote and HQ employees. Identifies if remote employees leave more frequently. Helps in evaluating work policy effectiveness.



3. Attrition by Gender

Insights:

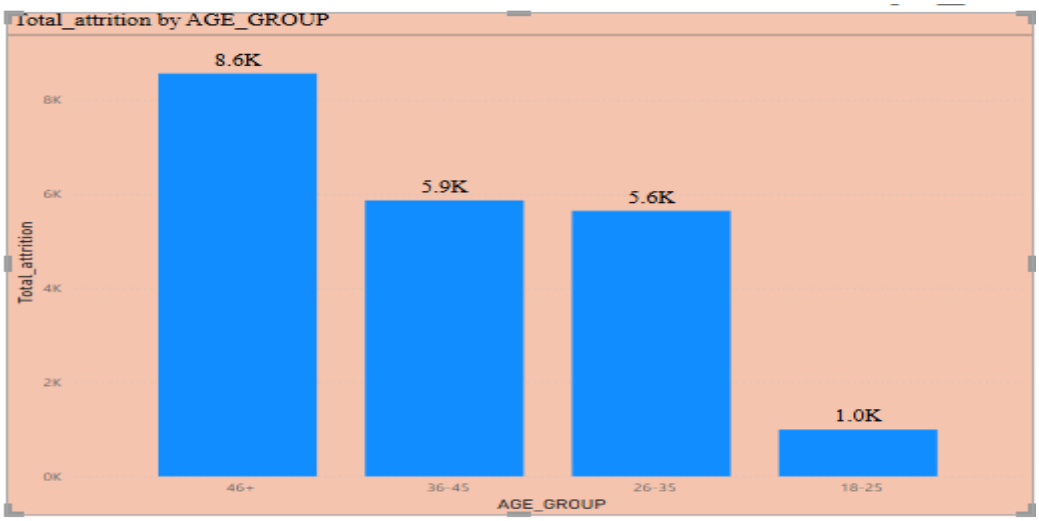
Shows gender-wise attrition distribution. Helps identify if attrition is biased toward a specific gender. Supports diversity and inclusion monitoring.



4. Attrition by Age Group

Insights:

The highest attrition is observed in the **46+ age group (8.6K)**, indicating that senior employees contribute significantly to overall workforce exits. In contrast, the **18–25 age group shows the lowest attrition (1.0K)**, suggesting comparatively better retention among younger employees.



5. Attrition Employee Details

Insights:

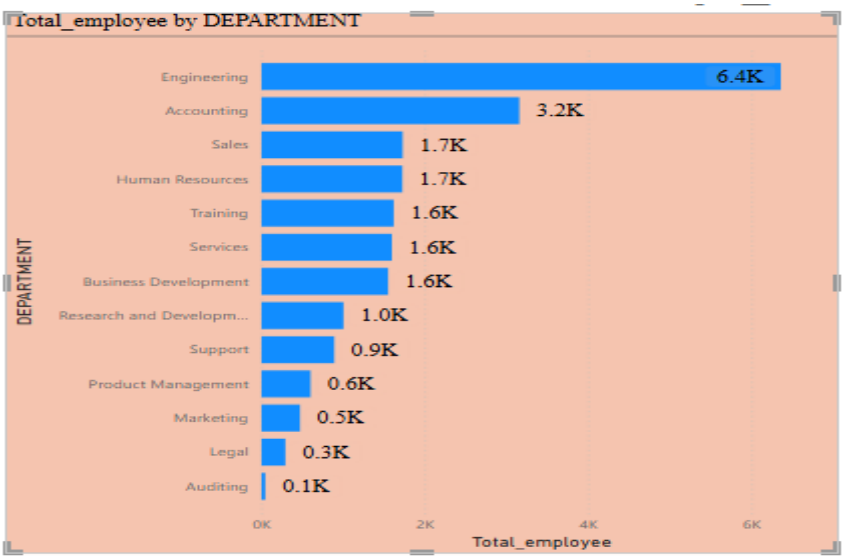
The table shows that each listed employee has an attrition value of **1**, indicating that these employee have exited the organization.

EMPLOYEE_ID	Total_attrition
00-0037846	1
00-0041533	1
00-0045747	1
00-0055274	1
00-0116166	1
00-0363185	1
00-0380704	1
00-0381660	1
00-0419202	1
00-0472287	1
00-0472832	1
00-0566380	1
00-0571075	1
00-0624189	1
00-0715212	1
00-0755645	1
00-0778934	1
00-0794247	1
00-0948136	1
00-0971612	1
00-1051096	1
00-1052230	1
00-1100714	1
00-1147503	1
00-1189819	1
Total	21075

6.Total employee by Department

Insights:

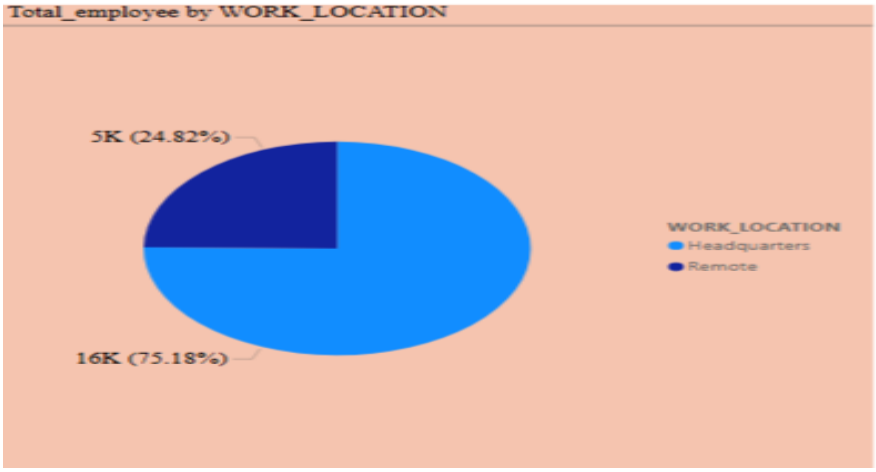
The chart shows the distribution of employees across different departments, highlighting which departments have the highest workforce concentration. Departments with larger employee counts may require stronger management focus and resource planning to maintain operational efficiency.



7.Total employee by work location

Insights:

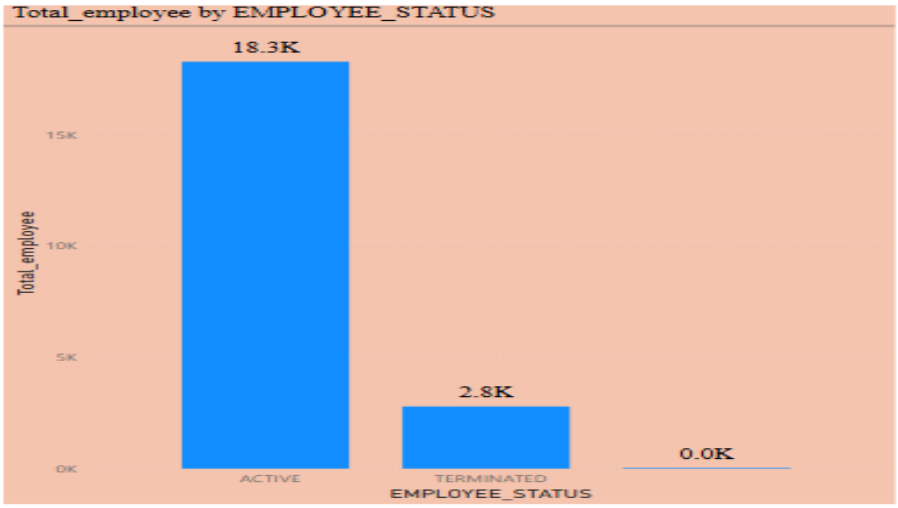
The workforce distribution shows how employees are split between Remote and Headquarters locations. This helps identify which work model dominates the organization and supports location-based workforce planning.



8.Total employee by Employee Status

Insights:

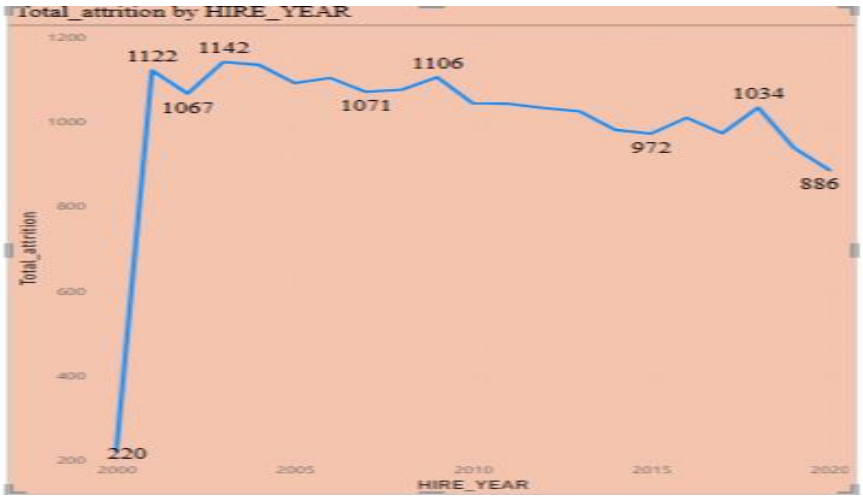
The majority of employees fall under the active category, while a smaller portion represents exited employees. This provides a quick view of overall workforce stability.



9.Total attrition by Hire Date

Insights:

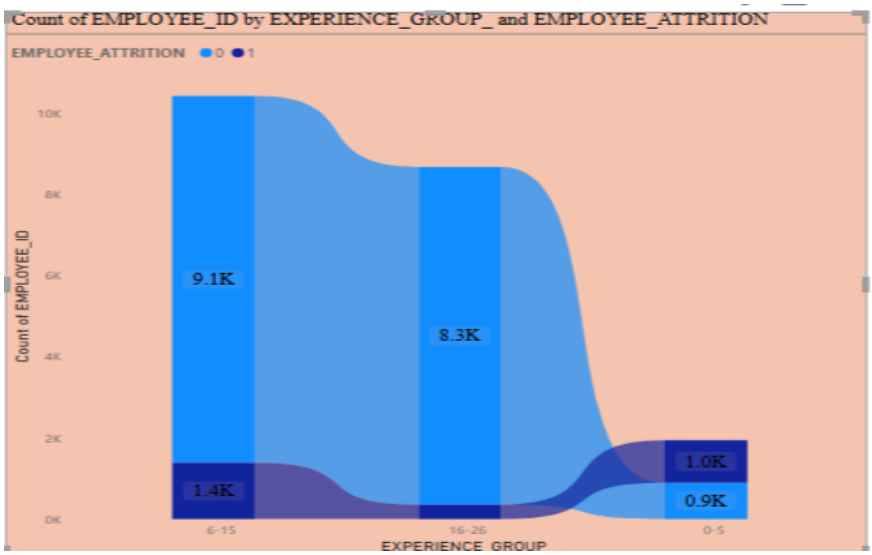
Attrition trends by hire date reveal which hiring periods experienced higher employee exits. This helps identify if certain recruitment batches had higher turnover rates.



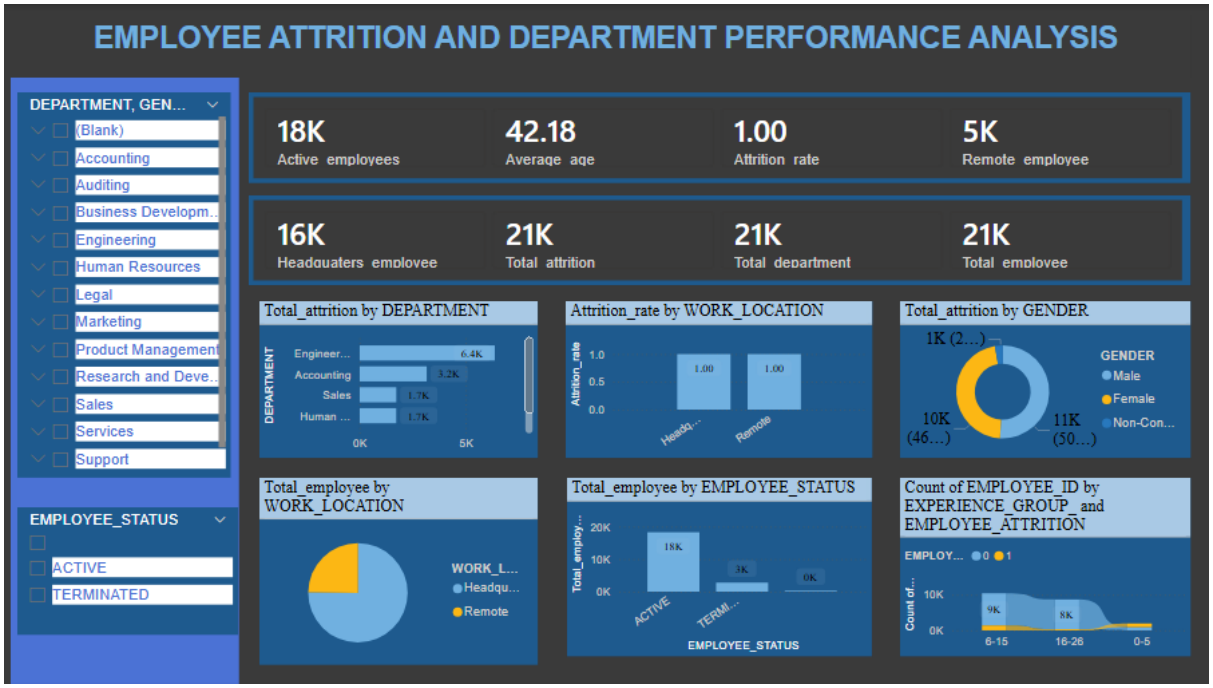
10.Count Of employee ID by Experience Group and Employee Attrition

Insights:

Attrition varies across experience groups, indicating that certain tenure levels contribute more to workforce exits. This insight supports targeted retention strategies based on employee experience levels.



Dashboards: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Business Developm...
Engineering
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Legal
Marketing
Product Management
Research and Deve...
Sales
Services
Support
EMPLOYEE_STATUS
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TERMINATED
18K
Active employees
42.18
Average age
1.00
Attrition rate
5K
Remote employee
16K
Headquaters employee
21K
Total attrition
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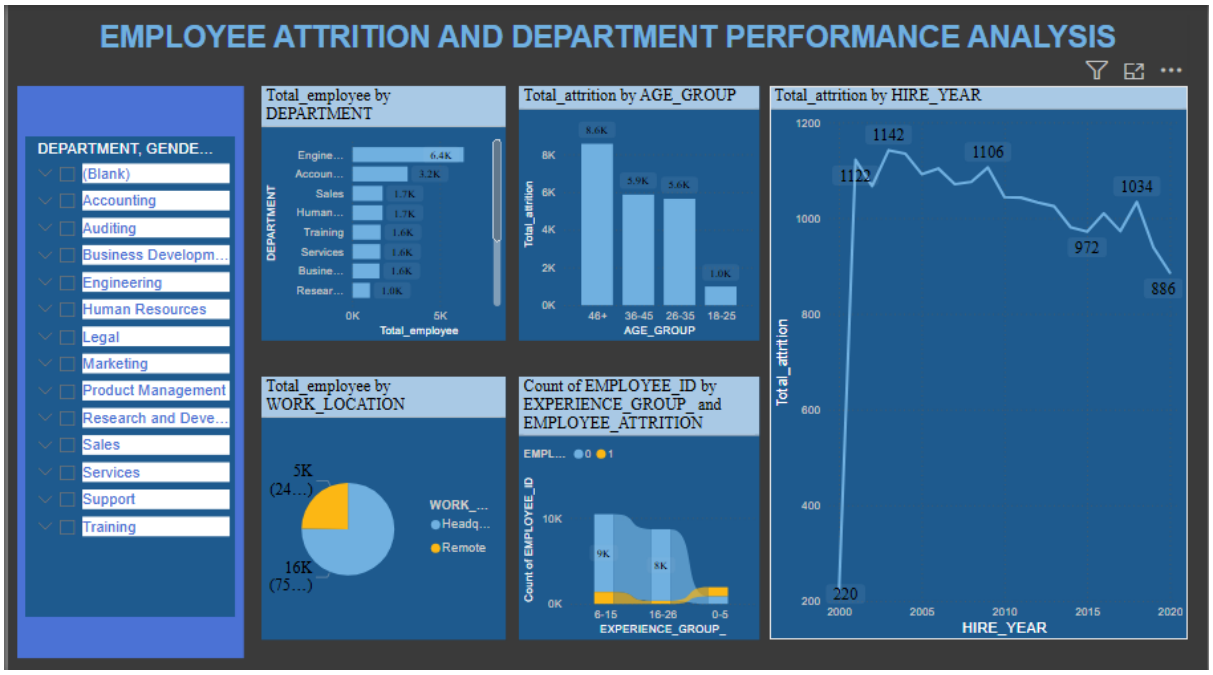
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Total_employee by DEPARTMENT
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Total_employee by WORK_LOCATION
Count of EMPLOYEE_ID by EXPERIENCE_GROUP_ and EMPLOYEE_ATTRITION
HIRE_YEAR

9. INSIGHTS:

- The overall attrition rate indicates a noticeable portion of employees leaving the organization, highlighting the importance of retention strategies.
- Certain departments show significantly higher attrition compared to others, suggesting possible workload imbalance or job dissatisfaction in those areas.
- Employees with lower years of experience show higher attrition levels, indicating that early-stage employees may require better onboarding and engagement programs.
- Workforce distribution across departments is uneven, which may impact productivity and operational efficiency.
- Remote and Headquarters employees show variation in attrition patterns, suggesting that work environment and flexibility may influence employee retention.
- Gender-wise attrition distribution appears balanced/unbalanced (adjust based on your dashboard result), providing insights into workforce diversity and stability.
- The majority of active employees fall within specific age groups, helping HR understand the demographic composition of the workforce.
- The dashboard highlights departments that require immediate HR attention to prevent future workforce shortages.

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10. CONCLUSIONS:

- The analysis reveals that employee attrition is significantly higher among senior age groups and specific experience levels, while younger employees show comparatively lower exit rates. Certain departments have a higher concentration of employees as well as higher attrition levels, indicating potential workload or retention challenges in those areas.
- The workforce distribution shows a stable number of active employees, which is a positive indicator of organizational strength. However, the noticeable attrition in particular departments, age groups, and tenure levels highlights areas requiring immediate HR attention.
- The findings suggest that employee retention strategies should focus on high-risk groups, especially experienced and senior employees. Overall, this project demonstrates how data driven insights can help organizations identify workforce trends, reduce attrition risk, and improve long-term employee stability.
- Higher attrition in certain groups = retention challenge

- Lower attrition in some groups = stable workforce
- Department imbalance = need for workforce planning
- Experience-based exits = need better engagement strategy